FINANCE I

Facultad de Economía y Negocios
Universidad de Chile

Prof.: Alejandro Bernales

Exam

Time Allowed: 1.5 Hours

Instructions:

The solemne has 3 questions, each of them have the same weight.

PLEASE WRITE YOUR NAME IN ALL THE PAGES.

PLEASE ONLY USE THE SPACE IN EACH QUESTION TO ANSWER ALL SECTIONS (USE ONLY ONE SIDE OF THE PAGE - ONE LANGUAGE)

The space provided is sufficient to give a concise and pertinent answer to each question and/or sub-point of a question.

Extra answer booklets will not be provided – please use the spaces provided only. Answers written outside the spaces or on other booklets will not be marked.

Electronic calculators may be used provided that they cannot store text

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Examen

Time Allowed: 1.5 Hours

Instrucciones:

El Control tiene 3 preguntas, cada una de ellas tienen la misma ponderación.

POR FAVOR ESCRIBA SU NOMBRE EN TODAS LAS HOJAS.

POR FAVOR UTILICE EL ESPACIO PROVISTO EN CADA PREGUNTA PARA RESPONDER CADA SECCIÓN (USAR UN SOLO LADO DE LA HOJA - UN SOLO LENGUAJE).

El espacio entregado es suficiente para entregar una respuesta concisa y pertinente a cada pregunta y/o sub-pregunta.

No se entregarán hojas adicionales para responder por favor utilice solamente el espacio entregado. Respuestas escritas fuera de este espacio o en otras hojas no serán revisadas.

Calculadoras electrónicas pueden ser utilizadas salvo las que puedan almacenar texto

Question I

Suppose that the risk premium on the market portfolio is 8% with a standard deviation of 22 %.

1a) What is the beta of a portfolio in which there is an allocation of 25% in Coca Cola, 20% in Apple, 25% in the S&P 500 and 30% in T-bills, if Coca Cola and Apple have betas of 1.10 and 1.25, respectively?, the portfolio is riskier or safer than the market portfolio (where the market portfolio is reflected in the S&P 500)?

Answer:

The portfolio β is:

$$\beta_p = w_{coca} * \beta_{coca} + w_{apple}\beta_{apple} + w_{S\&P500}\beta_{S\&P500} + w_{T-bill}\beta_{T-bill}$$

$$\beta_p = 0{,}25 * 1{,}10 + 0{,}20 * 1{,}25 + 0{,}251 + 0{,}3 * 0 = 0{,}775$$

Since 0,775 < 1, the portfolio has a lower level of risk than the market portfolio.

1b) The expected return on the portfolio described above is 10 %. Does the portfolio have a normal expected return based on the CAPM? when?

Answer:

Using CAPM if

$$\alpha = Expectedrate(r) - ExpectedratefromCAPM$$

- If $\alpha > 0$, the stock is underpriced (hence desirable to invest in).
- If $\alpha < 0$, the stock is overpriced (hence undesirable to invest in).

First, we have to calculate the Expected return of the portfolio, we use SML:

$$E(r_p) = r_f + \beta_p [E(r_m) - r_f]$$

We don't know rf, so:

$$E(r_p) - r_f = 0{,}775 * 8\% = 6{,}2\%$$

$$\alpha = Expectedrate(r) - E(r_p)$$

We must consider the rf.

$$\alpha = 10\% - r_f - 6{,}2\%$$

$$\alpha = 3{,}8\% - r_f$$

- It will depend on the level of r_f

Pregunta I

Suponga que la prima de riesgo en el portafolio de mercado es del 8% con una desviación estándar del 22%.

1a) ¿Cuál es la beta de un portafolio en el que hay invertido un 25% en Coca Cola, 20% en Apple, 25% en el S&P 500 y 30% en T-bills, si Coca Cola y Apple tienen betas de 1.10 y 1.25, respectivamente?, la cartera es más riesgosa o más segura que el portafolio de mercado (donde el portafolio de mercado se refleja en el S&P 500)?

Respuesta:

1b) La tasa de retorno esperada del portafolio descrito anteriormente es del 10%, ¿tiene un retorno esperado normal según el CAPM? cuando?

Respuesta:

1c) Coca cola has an expected return of 14,8% (and $\beta = 1{,}1$). Apple has expected return of 14% (and β=1,25). The expected return of the market portfolio is 13 %, and rf = 5%. What is the alpha of each stock?

Answer:

Now in this situation, we know the free rate ($r_f = 5\%$):

$$E(r_i) = r_f + \beta_i[E(r_m) - r_f]$$
$$E(r_{Coca}) = r_f + \beta_{Coca}[E(r_m) - r_f]$$
$$E(r_{Coca}) = 5\% + 1{,}1 * 8\% = 13{,}8\%$$

And Apples:

$$E(r_{Apple}) = 5\% + 1{,}25 * 8\% = 15\%$$

$$\alpha_{Coca} = 14{,}8\% - 13{,}8\% = 1\%,$$
$$\alpha_{Apple} = 14\% - 15\% = -1\%,$$

1d) The risk free rate is now 8%, you consider to invest in a fund with an asset allocation of 45% in Coca cola, 45% in Apple and 10% in T.bills, if Coca cola and Apple have betas of 1,10 and 1,25, respectively. What is the required rate o return on the project? (remember that the risk premium on the market portfolio is 8%)

Answer:

We must calculate again the portfolio's beta:

$$\beta_p = w_{coca}\beta_{coca} + w_{apple}\beta_{apple} + w_{T-bill}\beta_{T-bill}$$
$$\beta_p = 0{,}45 * 1{,}10 + 0{,}45 * 1{,}25 + 0{,}1 * 0 = 1{,}06$$
$$E(r_p) = r_f + \beta_p[E(r_m) - r_f]$$
$$E(r_p) = 8\% + 1{,}06 * 8\% = 16{,}48\%$$

1c) Coca cola tiene un retorno esperado del 14,8% (y β = 1,1). Apple tiene un retorno esperado del 14% (y β = 1,25). El retorno esperado del portafolio de mercado es del 13% y rf = 5%. ¿Cuál es el alfa de cada acción?

Respuesta:

1d) La tasa libre de riesgo ahora es del 8%, y usted considera invertir en un fondo que está distribuido con 45% en Coca Cola, 45% en Apple y 10% en T.bills, si Coca Cola y Apple tienen beta de 1,10 y 1,25, respectivamente. ¿Cuál es la tasa requerida de retorno del proyecto? (recuerde que la prima de riesgo en la cartera del mercado es del 8%)

Respuesta:

Question II:

2a) What are the seven basic characteristics that a good measure of risk must have (as discussed in class)?

Answer:

We only discussed 5 characteristics (not seven):

-It should consider the width of the return distribution
- It should give more weight to recent data
- It should consider the asymmetry of the return distribution
- It should consider the tails of the return distribution (extreme events)
- It should be easy to explain

2b) The expiration day for options that are traded on the Chicago Board Option Exchange (CBOE) is always the last trading day of the respective expiration month of the option contract. Why?

Answer

- That is false. As discussed in class, the expiration day is the third Friday of the expiration month.

Pregunta II:

2a) Cuales son las siete caracteristicas basicas que tiene que tener una buena medida de riesgo (segun lo discutido en clases)?

Respuesta

2b) El día de vencimiento de las opciones que se negocian en el Chicago Board Option Exchange (CBOE) es siempre el último día de negociación ("the last trading day") del respectivo mes de vencimiento del contrato de opción. Cual es la razón?

Respuesta

Question III:

A stock price is currently \$30. During each 2-month period it will increase by 8% or reduce by 10% (for the next 4 months). The risk-free interest rate is 5%. Use a two-step tree to calculate the value of a derivative that pays off in 4 months:

$$Payoff = [\max(30 - S_T, 0)]^2,$$

where S_T is the stock price in 4 months (i.e. an European-Style derivative). If the derivative is American-style, should it be exercised early?

Answer:

This type of option is known as a power option. A tree describing the behavior of the stock price is shown in Figure M . The risk-neutral probability of an up move, p, is given by

$$p = \frac{e^{0.05\times 2/12} - 0.9}{1.08 - 0.9} = 0.6020$$

Calculating the expected payoff and discounting, we obtain the value of the option as

$$[0.7056 \times 2 \times 0.6020 \times 0.3980 + 32.49 \times 0.3980^2]e^{-0.05\times 4/12} = 5.394$$

The value of the European option is 5.394. This can also be calculated by working back through the tree as shown in Figure M . The second number at each node is the value of the European option.

Early exercise at node C would give 9.0 which is less than 13.2449. The option should therefore not be exercised early if it is American.

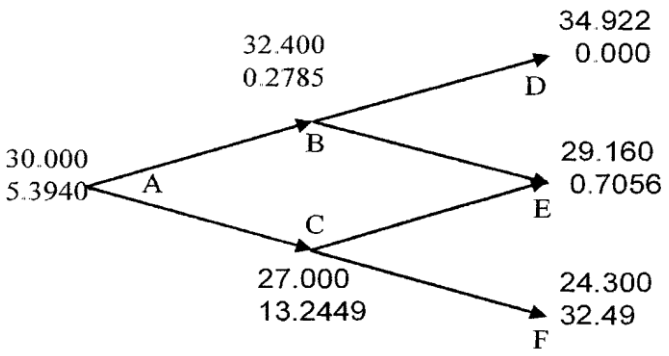


Figure M Tree to evaluate European power option
At each node, upper number is the stock price; next number is the option price

Pregunta III:

Una acción actualmente vale $30, la cual cada 2 meses subirá un 8% o decrecerá un 10% (lo que ocurrirá por los próximos 4 meses). La tasa libre de riesgo es de un 5% anual. Usando un árbol binomial de dos pasos, calcule el valor (al día de hoy) de un derivado que paga ("payoff") a la madurez en 4 meses:

$$Payoff = [\max(30 - S_T, 0)]^2,$$

donde S_T es el valor de la acción en 4 meses más (i.e. un derivado del estilo "Europeo"). ¿Si el derivado fuera del tipo de ejercicio "Americano", debiese ser ejercido anticipadamente?

Respuesta